

Firewall Rulebase Governance and Rule Register

Purpose: The audit-facing record of the pfSense firewall rulebase. It states the principles the rulebase is built on, records every rule with its business justification and the control it satisfies, and logs every change. It is written so that an auditor (or a new engineer) can understand *why* each rule exists without asking anyone.

Why this document exists: A firewall configuration on its own is not auditable. Auditors ask three questions of every rule: *why does it exist, who approved it, and is it still needed?* This register answers all three in one place. The discipline mirrors **PCI-DSS Requirement 1** (documented ruleset with justification and periodic review), applied here to a NIST 800-53 / CIS Controls context.

Status: Living document. Updated every time a rule is added, changed, or removed.

1. Rulebase principles

#	Principle	What it means in practice	Framework
1	Deny by default	Nothing passes unless a rule allows it. Every interface ends in an implicit deny.	NIST SC-7(5), CIS 4.4
2	Least privilege	Every rule is the narrowest possible: specific source, destination, service. No <code>any</code> → <code>any</code> .	NIST CM-7, AC-6, CIS 4.8
3	Documented justification	Every rule has a recorded business reason, owner, and date (this register).	NIST CM-2/CM-6, PCI-DSS 1.1
4	Correct ordering	Rules are ordered so specific precedes broad; no shadowed or redundant rules.	NIST CM-6
5	Logging & accountability	Denies and sensitive allows are logged to the SIEM (Wazuh).	NIST AU-2/AU-3
6	Change control	Changes are proposed, approved, dated, reversible (disable before delete).	NIST CM-3
7	Periodic recertification	The rulebase is reviewed on a fixed cadence; stale rules removed.	NIST CA-7, PCI-DSS 1.1.7
8	Boundary enforcement	Rules enforce the VLAN/zone boundaries defined by segmentation.	NIST SC-7, CIS 12

2. Rule register

Each rule carries a stable **Rule ID** (`<TAB>-<NN>`) that does not change even if the rule's position moves. "Log" = whether the rule logs matches to the SIEM. "Status" = Built or Planned.

2.1 MANAGEMENT interface (VLAN 10, source = admin/control plane)

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
MGMT-01	MANAGEMENT net	This Firewall	PFSense_MGMT (443,80,22)	Pass	Yes	Anti-lockout: guarantee admin access to pfSense from the management plane before broad rules are removed. Admin access is logged for accountability.	AC-17, CM-3, AU-2	Built
MGMT-02	MANAGEMENT net	This Firewall	TCP/UDP 53	Pass	No	DNS resolution via the pfSense resolver for management hosts.	SC-7	Built
MGMT-03	MANAGEMENT net	This Firewall	UDP 123	Pass	No	Time sync (NTP) from pfSense; correct time is required for logging and Kerberos.	AU-8	Built
MGMT-04	MANAGEMENT net	any	ICMP echo-request	Pass	No	Diagnostic reachability (ping) for troubleshooting. Echo-request only (least privilege); replies return via stateful inspection.	CA-7	Built
MGMT-05	MANAGEMENT net	SRV1_DC	AD_TCP	Pass	No	Domain membership (TCP) for management-plane Windows hosts joining ad.biira.online .	AC-4	Built
MGMT-06	MANAGEMENT net	SRV1_DC	AD_UDP	Pass	No	Domain membership (UDP).	AC-4	Built
MGMT-07	MANAGEMENT net	SRV1_DC	AD_RPC_DYNAMIC	Pass	No	RPC high ports for domain join, Group Policy, replication.	AC-4	Built
MGMT-08	MANAGEMENT net	SRV1_DC	MGMT_TCP	Pass	Yes	Administrative access (RDP, WinRM/Ansible) to the DC. Logged as privileged access.	AC-6, AC-17, AU-2	Built

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
MGMT-09	MANAGEMENT net	LAB_NETS	any	Pass	No	Administrative reach into all lab VLANs from the management plane (explicit; to be tightened per-service over time).	AC-6	Planned
MGMT-10	MANAGEMENT net	NOT LAB_NETS	any	Pass	No	Internet access for the management plane (destination inverted = anywhere except the lab).	SC-7	Planned
(implicit)	any	any	any	Deny	-	Default deny. Anything not explicitly allowed is dropped.	SC-7(5)	Built-in

2.2 ENTERPRISELAN interface (VLAN 50, source = the domain controller's segment)

This tab governs what VLAN-50 hosts (currently only `DC01`) may *initiate*. Inbound access to the DC is governed by the source VLANs' tabs (e.g. VLAN 10 via MGMT-05-08) and replies are stateful, so here we permit only what the DC legitimately originates and deny lateral movement into other VLANs.

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
ENT-01	ENTERPRISELAN net	This Firewall	TCP/UDP 53	Pass	No	DNS resolution/forwarding via pfSense.	SC-7	Built
ENT-02	ENTERPRISELAN net	This Firewall	UDP 123	Pass	No	NTP time sync; correct time underpins Kerberos and logging.	AU-8	Built
ENT-03	ENTERPRISELAN net	any	ICMP echo-request	Pass	No	Diagnostic reachability (outbound ping).	CA-7	Built
ENT-04	ENTERPRISELAN net	NOT LAB_NETS	any	Pass	No	Internet access (updates, activation, CRL, public DNS). Excludes all lab VLANs.	SC-7	Built
(implicit)	any	any	any	Deny	-	Default deny. VLAN 50 cannot initiate into other VLANs or reach pfSense admin.	SC-7(5), AC-4	Built-in

The `any→any` rule that previously governed this interface was deleted 2026-08-30 after validation.

2.3 REDTEAM interface (VLAN 30, the attack segment)

VLAN 30 is treated as **untrusted**. The baseline contains it from the entire internal lab: the attack box may function (name resolution, time, internet for tooling) but has **no standing path to the domain controller, any other VLAN, or the pfSense admin interface**. Access to a target for an actual exercise is granted as a temporary, specific rule

and removed afterwards, so a compromised or careless attack box cannot roam the network.

Rule ID	Source	Destination	Service	Action	Log	Justification	Control	Status
RED-01	REDTEAM net	This Firewall	TCP/UDP 53	Pass	No	Name resolution so the attack host functions.	SC-7	Built
RED-02	REDTEAM net	This Firewall	UDP 123	Pass	No	Time sync; accurate timestamps for exercise evidence.	AU-8	Built
RED-03	REDTEAM net	NOT LAB_NETS	any	Pass	No	Internet only, for tooling and updates. Excludes every lab VLAN.	SC-7	Built
(implicit)	any	any	any	Deny	,	Default deny. No path to DC01, other VLANs, or pfSense admin.	SC-7(5), AC-4	Built-in

Deliberate omission: unlike MANAGEMENT and ENTERPRISELAN, there is **no ICMP-to-any rule** here. An outbound ping rule would let the attack box sweep the internal VLANs for live hosts, which is exactly the reconnaissance step to deny. ICMP to the internet still works through RED-03. The absence of a rule is itself the control.

The screenshot displays the pfSense web interface for the Firewall Rules configuration. The 'REDTEAM' tab is selected, showing a list of rules. A green message at the top states: 'The changes have been applied successfully. The firewall rules are now reloading in the background. Monitor the filter reload progress.' Below this, the 'REDTEAM' tab is active, showing four rules:

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/0 B	IPv4	TCP/UDP	REDTEAM subnets	*	This Firewall (self)	53 (DNS)	*	none	RED-01 - DNS to pfSense resolver. NIST SC-7
☒	0/0 B	IPv4	UDP	REDTEAM subnets	*	This Firewall (self)	123 (NTP)	*	none	RED-02 - NTP time sync from pfSense. NIST AU-8
☒	0/0 B	IPv4 *	*	REDTEAM subnets	*	! LAB_NETS	*	*	none	RED-03 - Internet only (everything except the lab). NIST SC-7
☒	0/0 B	IPv4 *	*	REDTEAM subnets	*	*	*	*	none	RedTeam Firewall

At the bottom of the rules list, there are buttons for 'Add', 'Add', 'Delete', 'Toggle', 'Copy', 'Save', and 'Separator'. The footer of the page indicates 'pfSense is developed and maintained by Netgate. © ESF 2004 - 2026 View license.'

Figure 13.8: The REDTEAM tab. RED-01 to RED-03 are active; the legacy RedTeam Firewall any-to-any rule is disabled (greyed) pending deletion. No rule permits VLAN 30 to reach any lab subnet.

Validation status: deferred. There is currently no host on VLAN 30 (the Kali VM has not been built yet), so the containment test cannot be run. Once Kali exists, run from it: `curl -I https://google.com` (expect success, RED-03), `ping -c 2 192.168.50.2` and `nc -zv 192.168.50.2 443` (expect both to fail, proving containment from DC01). Record the result in section 3.1.

Remaining tabs (BLUETEAM, DEVOPS, MONITORING) are registered here as they are built. Their target layouts are in docs/11-domain-controller-firewall.md section 6.

3. Change control log

Every change to the rulebase is recorded here: what changed, when, who made it, whether it was tested, and how it can be rolled back.

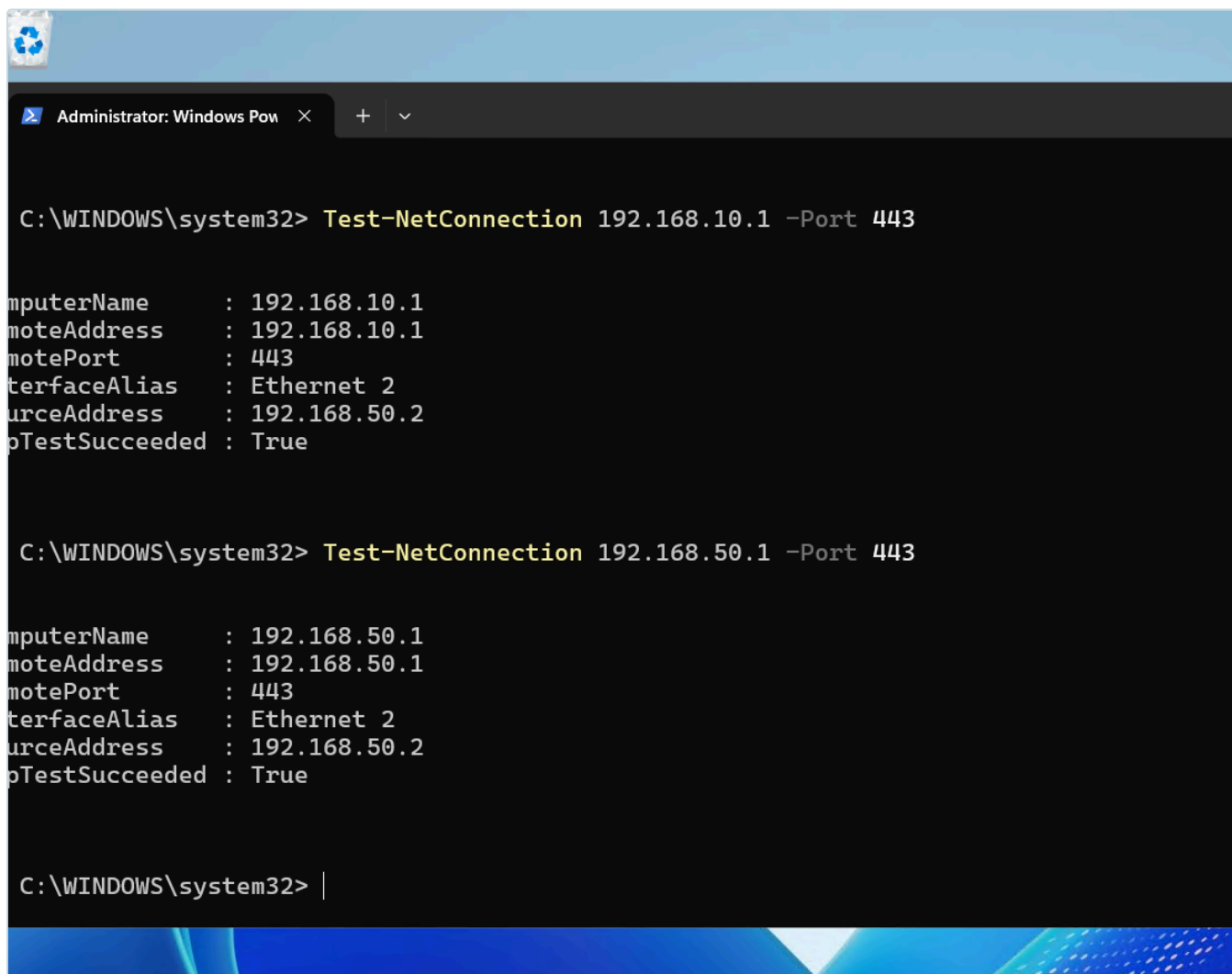
Date	Rule(s)	Change	By	Tested	Rollback
2026-08-15	Aliases	Created SRV1_DC , AD_TCP , AD_UDP , AD_RPC_DYNAMIC , MGMT_TCP , LAB_NETS	Noble Antwi	Verified in Aliases UI	Delete aliases (no rules depend yet)
2026-08-15	MGMT-01	Added anti-lockout rule at top of MANAGEMENT	Noble Antwi	Reload pfSense UI from laptop, confirmed access	Disable rule; broad "All" rule still present
2026-08-15	MGMT-02	Added DNS-to-pfSense rule below MGMT-01	Noble Antwi	UI reachable after apply	Disable rule
2026-08-15	MGMT-03	Added NTP-to-pfSense rule	Noble Antwi	UI reachable after apply	Disable rule
2026-08-15	MGMT-04	Added ICMP rule; corrected subtype echo-reply → echo-request (least privilege, outbound ping)	Noble Antwi	Rule applied	Disable rule
2026-08-15	MGMT-05 to 08	Added the four DC-access rules (AD_TCP, AD_UDP, AD_RPC_DYNAMIC, MGMT_TCP) to SRV1_DC; MGMT-08 logging enabled	Noble Antwi	Aliases resolve as links; applied	Disable rules; DC still reachable via broad "All" until it is removed
2026-08-30	ENT-01 to 04	Built ENTERPRISELAN outbound rules (DNS/NTP to pfSense, ICMP, internet via !LAB_NETS); deleted the prior any→any	Noble Antwi	Isolation test flipped True→False after a manual filter reload (see 3.1 and incident below)	Re-add a temporary any→any on ENTERPRISELAN
2026-09-03	RED-01 to 03	Built REDTEAM baseline (DNS/NTP to pfSense, internet via !LAB_NETS); no ICMP-to-any by design; legacy any-to-any disabled pending deletion	Noble Antwi	Not yet: no host on VLAN 30 (Kali not built). Test deferred, see 2.3	Re-enable the legacy any-to-any rule
2026-08-30	(incident)	Stale filter reload , GUI rule changes were saved but the kernel kept enforcing the old ruleset, so isolation tests kept passing. Root-caused by systematic elimination (floating rules empty, anti-lockout disabled, packet-filtering enabled, any→any deleted, none had effect). Resolved via Status → Filter Reload → Reload Filter , which loaded the current ruleset.	Noble Antwi	Both isolation tests then returned False	N/A (diagnostic)

3.1 Control validation tests

Evidence that a control does what its policy claims. Each test is repeatable, and its result is recorded before and after the change that is meant to affect it.

Date	Control	Test	From → To	Expected	Result
2026-08-15	Management-plane isolation (baseline)	pfSense web UI load	Admin laptop (VLAN 10) → pfSense admin	Reachable	True
2026-08-15	Management-plane isolation (baseline)	Test-NetConnection 192.168.50.1 -Port 443	DC01 (VLAN 50) → pfSense admin 443	Reachable now (VLAN 50 not yet tightened)	True (baseline)
2026-08-30	Management-plane isolation (enforced)	Test-NetConnection 192.168.50.1 -Port 443	DC01 (VLAN 50) → pfSense admin 443	Blocked after ENTERPRISELAN tightened	False ✓
2026-08-30	Management-plane isolation (cross-VLAN)	Test-NetConnection 192.168.10.1 -Port 443	DC01 (VLAN 50) → Management gateway admin 443	Blocked	False ✓

The change from **True to False**, while `dcdiag`, DNS and internet still pass, is the proof that VLAN 50 is contained: it can use the gateway's DNS/NTP and reach the internet, but cannot reach the pfSense admin UI on any interface, nor pivot into other VLANs. (Ping still succeeds, because ENT-03 permits ICMP, showing the rule is surgical, not a blanket block.)



```

Administrator: Windows PowerShell
C:\WINDOWS\system32> Test-NetConnection 192.168.10.1 -Port 443

ComputerName      : 192.168.10.1
RemoteAddress     : 192.168.10.1
RemotePort        : 443
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.50.2
PingTestSucceeded : True

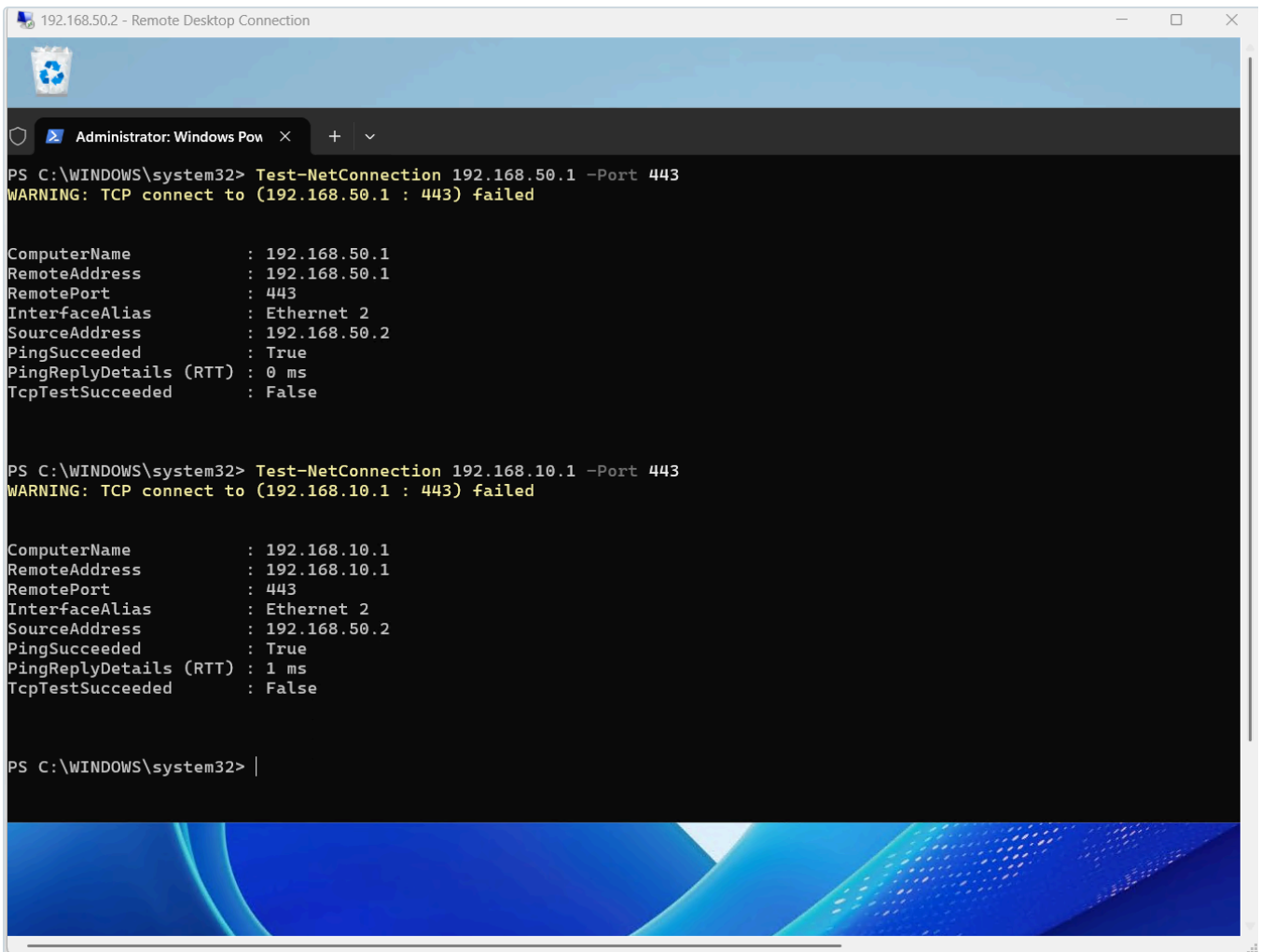
C:\WINDOWS\system32> Test-NetConnection 192.168.50.1 -Port 443

ComputerName      : 192.168.50.1
RemoteAddress     : 192.168.50.1
RemotePort        : 443
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.50.2
PingTestSucceeded : True

C:\WINDOWS\system32>

```

Figure 13.5: Before: both `Test-NetConnection ... 443` return `True`. The DC could reach the pfSense admin UI on its own VLAN and cross-VLAN.



```
192.168.50.2 - Remote Desktop Connection

Administrator: Windows Pow x + v

PS C:\WINDOWS\system32> Test-NetConnection 192.168.50.1 -Port 443
WARNING: TCP connect to (192.168.50.1 : 443) failed

ComputerName      : 192.168.50.1
RemoteAddress     : 192.168.50.1
RemotePort        : 443
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.50.2
PingSucceeded     : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False

PS C:\WINDOWS\system32> Test-NetConnection 192.168.10.1 -Port 443
WARNING: TCP connect to (192.168.10.1 : 443) failed

ComputerName      : 192.168.10.1
RemoteAddress     : 192.168.10.1
RemotePort        : 443
InterfaceAlias    : Ethernet 2
SourceAddress     : 192.168.50.2
PingSucceeded     : True
PingReplyDetails (RTT) : 1 ms
TcpTestSucceeded  : False

PS C:\WINDOWS\system32> |
```

Figure 13.6: After: both return False while Ping stays True. VLAN 50 is denied admin access on every interface, yet the DC's own services keep working.

Operational note, the stale filter reload. The isolation test kept returning True even after the rules were correct, because pfSense had not loaded the new ruleset into the kernel: the GUI and the running filter were out of sync. Systematic elimination (floating rules, anti-lockout, packet-filtering, deleting the rule) confirmed no rule change was taking effect. It was resolved with **Status → Filter Reload → Reload Filter**. **Lesson:** after tightening rules, if behaviour does not change, verify the filter actually reloaded, a stuck reload silently enforces the old rules and can make a control look broken when it is merely not loaded.

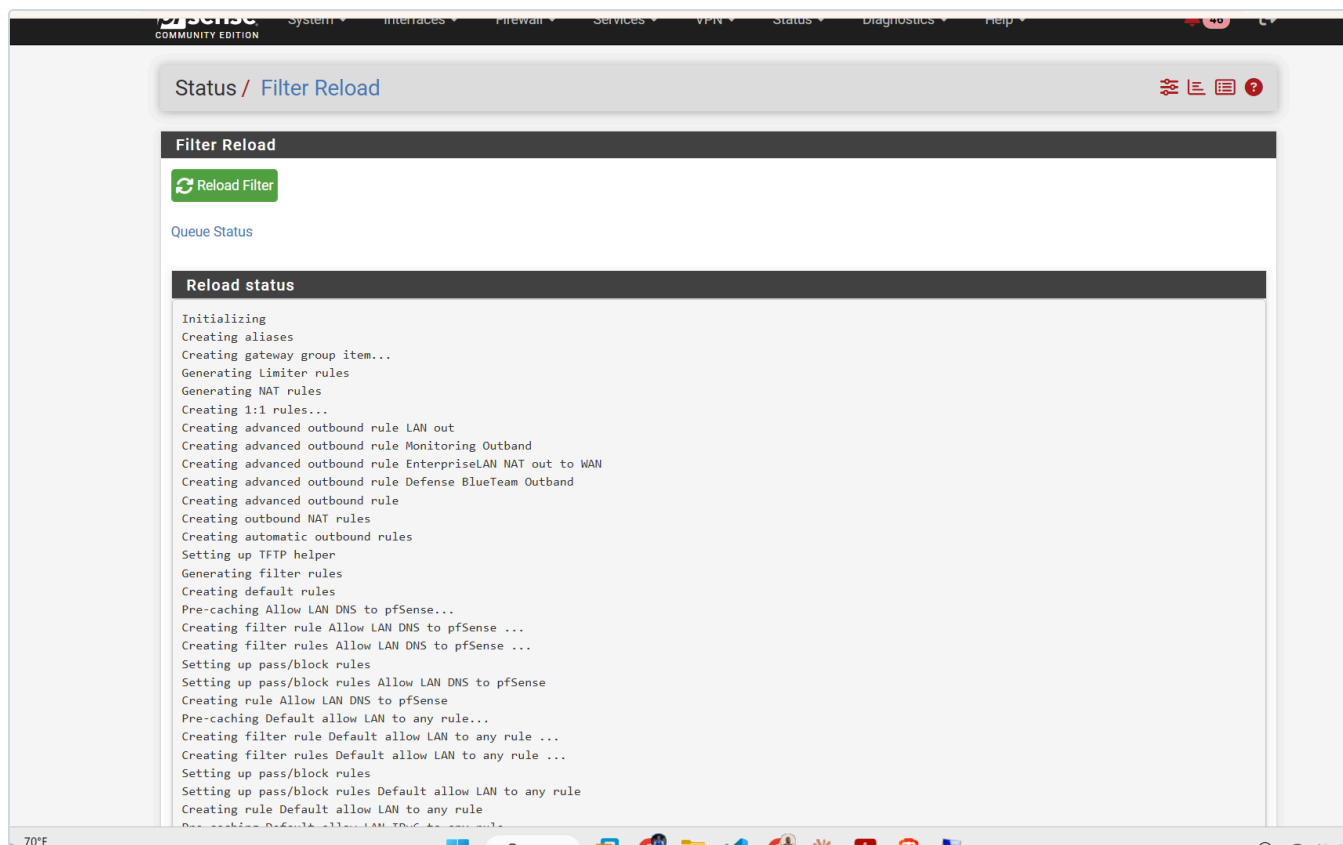


Figure 13.7: Status → Filter Reload. Manually reloading compiled and loaded the current ruleset, after which the isolation control took effect.

4. Review and recertification

- **Cadence:** the full rulebase is reviewed every **6 months** (PCI-DSS 1.1.7 discipline), or after any significant architecture change (new VLAN, new server role).
- **What a review checks:** each rule still has a valid justification; no shadowed/redundant/stale rules; least-privilege still holds; logging still appropriate; the register matches the live config.
- **Evidence:** review dates and outcomes are appended to the change control log.
- **Next review due:** 2027-02-15.

5. How to read a rule in this register (worked example: MGMT-01)

MGMT-01, Source MANAGEMENT net, Destination This Firewall, Service PFSENSE_MGMT (443/80/22), Action Pass, Log Yes.

- It only matches traffic **from** the management VLAN **to** pfSense itself, and only on the admin service ports. A host on any other VLAN, or aimed at any other destination, does not match it.
- It is logged because administrative access to the firewall is privileged activity that must be attributable.
- Its justification (anti-lockout) explains *why* it must be the first rule: if a later change removes the broad "allow" rule, this one guarantees the administrator can still reach pfSense to fix any mistake.

- Its controls (AC-17 remote access, CM-3 change control, AU-2 auditable events) are the standards language an auditor maps it to.

That is the level of "why" every rule in this register carries.

6. Known hardening opportunities (consciously accepted for now)

Recording a known weakness, the reason it is accepted, and the planned fix is itself a control (NIST CA-5, Plan of Action and Milestones). It shows a risk was identified and decided on deliberately, not missed.

ID	Observation	Risk	Current decision	Planned hardening	Trigger
H-01	MGMT-08 (RDP/WinRM to the DC) uses source = MANAGEMENT subnet , not specific hosts. Any device on VLAN 10 can <i>reach</i> the DC admin ports (network-location trust).	A rogue/compromised device placed on VLAN 10 could reach admin services. Mitigated by: (a) the DC still requires valid credentials + Remote Desktop Users membership to log in; (b) VLAN 10 is physically/administratively controlled.	Accepted while the lab has a single controlled admin laptop.	Create an <code>ADMIN_HOSTS</code> alias (specific admin/PAW IPs) and change MGMT-08 source to it; add MFA on the DC; move admin to a PAW.	When a second machine joins VLAN 10, or the PAW is built (see <code>docs/12</code>).

| H-02 | The **TAILSCALE** interface has broad rules: any tailnet device → pfSense admin; SSH to any destination; and `192.168.0.0/16` → any (reach every VLAN). Remote access therefore bypasses the internal VLAN segmentation. | A compromised or shared tailnet device would have full lab reach, including the RedTeam VLAN. | **Accepted**, Tailscale is WireGuard-authenticated and the tailnet holds only 4 owner-controlled devices (pfSense subnet router `noble-homelab`; admin laptop `noble-host` `100.118.195.0`; `lab-devops-svc01`; `iphone182`). It is the sole remote-admin path, so tightening it hastily risks lockout. | Scope the pfSense-admin rule to the admin device (`noble-host`); optionally exclude RedTeam (VLAN 30) from Tailscale reach; prefer Tailscale ACLs (admin console) for device-level control. Physical VLAN-10 access (MGMT-01) is the fallback. | When convenient, and **before adding any non-owner device** to the tailnet. |

This register grows as other rules or segments are found to have similar location-based trust that could later be tightened to host- or identity-based trust (the Zero Trust direction, NIST SP 800-207).